Kathmandu College of Technology

Lokanthali, Bhaktapur

Lab report on

Computer fundamentals and applications (CACS-101}

As required for the partial fulfillment of

BCA 1st Semester

Affiliated to Tribhuvan university

Submitted by

Smriti Shrestha

Roll no: 25

Submitted to

Department of BCA

CERTIFICATE FROM THE SUPERVISOR

I hereby affirm that the Lab Report titled "**Computer Fundamentals and Application**" is an academic undertaking carried out by **'SMRITI SHRESTHA**' as part of the partial fulfillment of the requirements for the degree of **Bachelor of Computer Application** at the esteemed Faculty of Humanities and Social Sciences, Tribhuvan University, under my unwavering guidance and supervision. To the best of my discernment, the content presented in the Lab Report is the original work of **'SMRITI SHRESTHA'** and reflects her individual effort and ingenuity in the field.

Signature of the Supervisor: -

Name: -

Designation: -

Date: -

ACKNOWLEDGEMENT

I would like to extend my heartfelt appreciation to **Prajwol sir** and the entire faculty of Humanities and Social Sciences for providing me with the opportunity to complete this project. I am immensely grateful for the support and guidance received during the course of my studies.

The requirement to submit this report stems from the guidelines set forth by **Kathmandu College of Technology (KCT)**, where it is mandatory to obtain practical marks. Our esteemed teacher for Computer Fundamentals and Application has been a constant source of motivation, providing valuable insights on how to structure and prepare the report to achieve optimal results. I am particularly grateful to all my fellow BCA students who were actively engaged in this endeavor.

This report is intended to be a valuable resource for future reference, and I am committed to delivering a high-quality output in accordance with the academic standards set forth by KCT.

Thank You

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A. Word Processing Software

1. Write an application to apply the post of Java programmer with your bio-data.

Madhyapur thimi-4, Bhaktapur

Province no: 1

2079/12/03

Mr. Jack Mullin

Oracle Corporation

Oracle parkway

Redwood shores, ca 94065

Sub: Application for post of Graphic Designer

Dear Sir,

I am very interested in the position of Graphic Designer at your esteemed organization. I came across the position via your advertisement in New York Times.

I believe that I can give my best for your company; I am confident that I have the skills to excel at your dynamic organization. I have always dreamed of being a part of your company due to the development opportunities you provide. I request you to give me an opportunity to learn and grow at your esteemed firm through this job role.

I am confident that my skills perfectly match your requirements. I have also attached a copy of CV with this application,

Thank you for considering my application. I hope you find it suitable so that we can arrange a meeting. I look forward to hearing from you.

Yours sincerely,
Smriti Shrestha
9818986530
Smritishrestha058@gmail.com

Smriti Shrestha

Bhaktapur, Nepal | 9823525425 | Email: smritishrestha058@gmail.com

Objective

To pursue my career in an organization with total effort and devotion with an ambition to rise to the top management where my abundant skills and full academic potential would lay a foundation for my career.

Education

2022-present: Bachelor in computer Application
Field of study: BCA
Kathmandu College of Technology

2020-2022: Plus 2
Field of Study: Management
Supreme Academy H.S.S

Experience and Internship

R.D Auto Trade

Service Department Operator

Responsibilities:

- Customer Handling
- Cash Management
- Stock Management

Skills

- Excellent Written and verbal communication skills
- Excellent MS-Office knowledge
- Can handle the work pressure and can work in any kind of scenario as suggested

Declaration

All over information has been completed by me and is accurate.

2. Prepare any type of form to fill up user's detail in any word processing software.

Kathmandu College of Technology

(Formerly known as Samajik College)

Reg no _____

ADMISSION FORM

Paste you recent Password size photo here

For office use:

Course: _____

Admission Date: _____

1.Name:

2. Father's Name:

3. Gender: Male Female

4. Date of Birth:

5. Phone no:

6. Nationality:

7. Address:

Province

District

Municipality

Ward No

_____ Principal's Signature

_____ Student's Signature

3. Prepare a two months' calendar using table.

Baishakh 2080						April/May
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			1 16	2 17	3 18	4 19
5 20	6 21	7 22	8 23	9 24	10 25	11 26
12 27	13 28	14 29	15 30	16 1	17 2	18 3
19 4	20 5	21 6	22 7	23 8	24 9	25 10
26 11	27 12	28 13				

Holidays: -

19 February: Democracy Day

Jestha 2080						May/June
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			1 14	2 15	3 16	4 17
5 18	6 19	7 20	8 21	9 22	10 23	11 24
12 25	13 26	14 27	15 28	16 29	17 30	18 1
19 2	20 3	21 4	22 5	23 6	24 7	25 8
26 9	27 10	28 11	29 12	30 13	31 14	

Holidays: -

21 March: Ghode Jatra

6 March: Holi

8 March: Women's Day

4. Design a well-formatted newspaper page containing text, pictures, drop cap, charts, etc.

THE KATHMANDU POST

Tuesday, April 11, 2023

A singer's untold story

Musician Arjun Kaushal opens up about his unstable career and the many challenges he has overcome on his journey. Arjun Kaushal has been active in the music industry for over three decades, but the composer, lyricist and singer still finds it difficult to sustain himself in the field. Kaushal isn't a formally trained musician and having worked so long in the music industry, he feels formal training isn't necessary for people who want to pursue a career in it. "At the beginning of my career, I just knew the eight notes—'Sa Re Ga Ma'. But that didn't stop me. I immersed myself in the field and picked up skills through regular practice," he says.

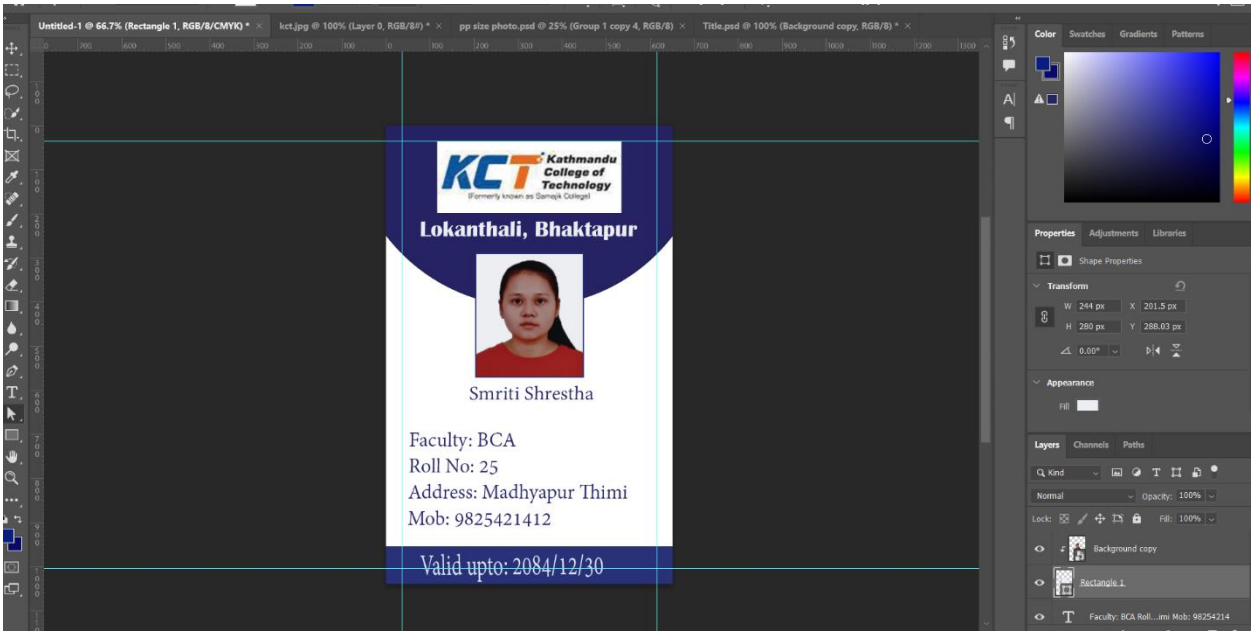
Will UML help the coalition on transitional justice bill?

As the ruling parties want to fast-track the bill, the role of the second-largest party is being keenly watched. Seventeen years have passed since the peace process began following the signing of the Comprehensive Peace Accord in 2006. But major political parties are still blaming each other for the delay in concluding the process. Needed amendments to the laws have not been made while the two commissions created to provide justice to the insurgency victims remain largely ineffective?

5. Design an attractive advertisements using Photoshop.

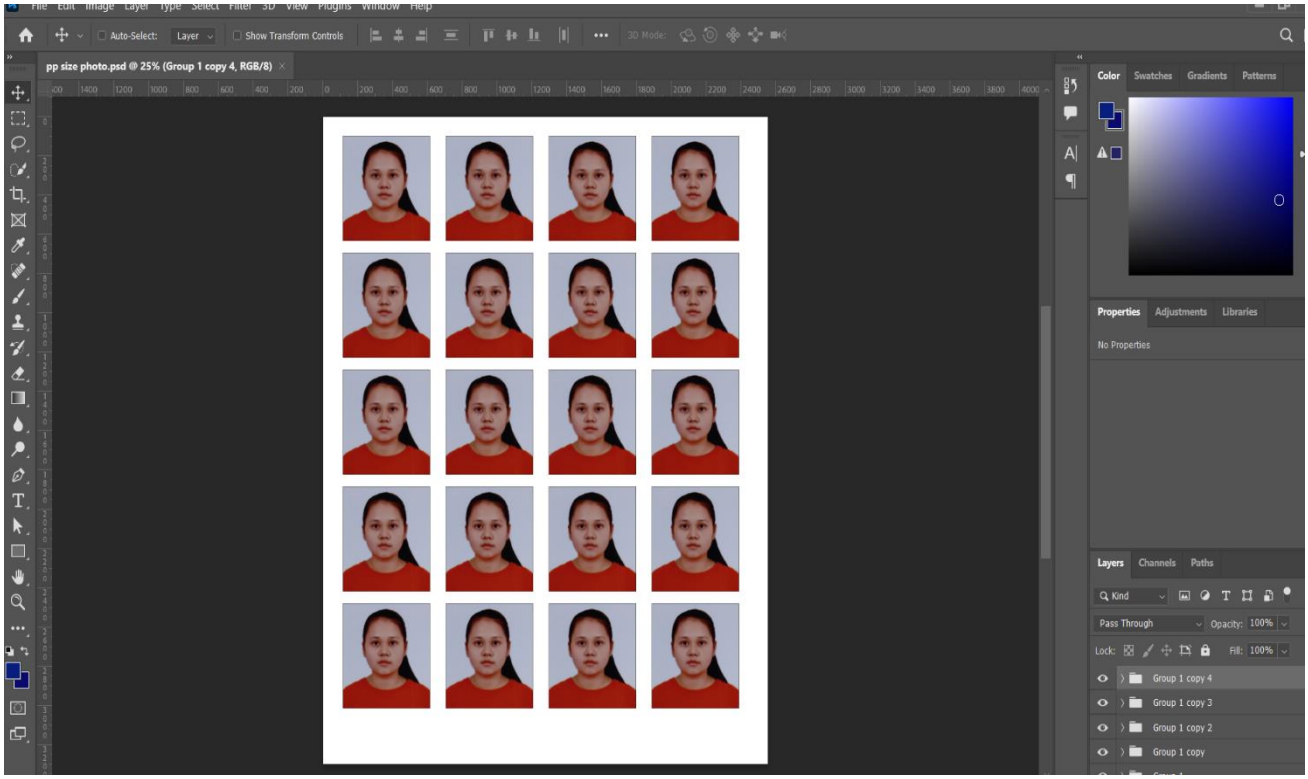
Final Result

6. Design an ID card for your college using Photoshop.



Final Result:

7. Prepare your passport size photo with white background in Photoshop.



Final Result:

B. Spreadsheet software

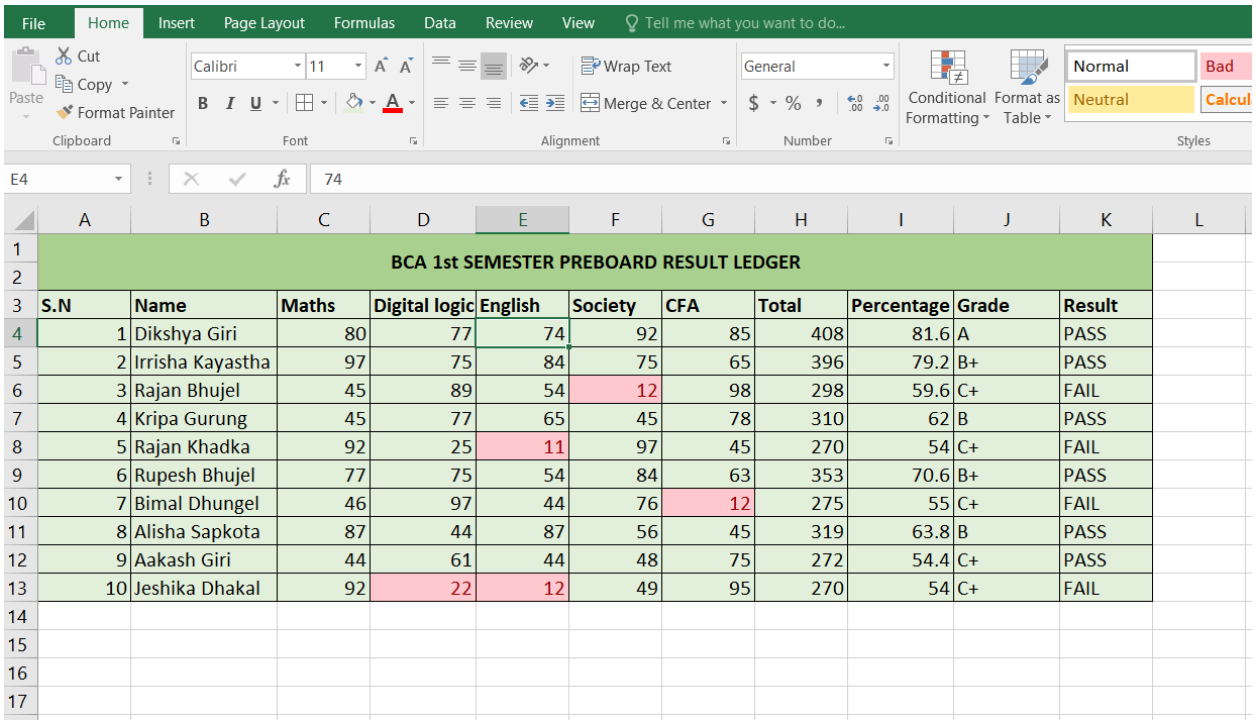
8. Create a marks ledger of 10 students as per the following conditions:

a) Calculate total marks

b) Calculate result, whether the student has passed or failed the examination.

c) Calculate the percentage of each student.

d) Calculate the grade of each student where 90-100% is A+, 80-90% is A, 70-80% is B+, 60-70% is B, 50-60% is C+, 40-50% is C and <40 is non-graded that is “NULL”.



BCA 1st SEMESTER PREBOARD RESULT LEDGER										
S.N	Name	Maths	Digital logic	English	Society	CFA	Total	Percentage	Grade	Result
1	Dikshya Giri	80	77	74	92	85	408	81.6	A	PASS
2	Irrisha Kayastha	97	75	84	75	65	396	79.2	B+	PASS
3	Rajan Bhujel	45	89	54	12	98	298	59.6	C+	FAIL
4	Kripa Gurung	45	77	65	45	78	310	62	B	PASS
5	Rajan Khadka	92	25	11	97	45	270	54	C+	FAIL
6	Rupesh Bhujel	77	75	54	84	63	353	70.6	B+	PASS
7	Bimal Dhungel	46	97	44	76	12	275	55	C+	FAIL
8	Alisha Sapkota	87	44	87	56	45	319	63.8	B	PASS
9	Aakash Giri	44	61	44	48	75	272	54.4	C+	PASS
10	Jeshika Dhakal	92	22	12	49	95	270	54	C+	FAIL

Formulas used:

Total: =SUM (C4:G4)

Percentage: =H4/5

Grade: =IF(I4>=90,"A+",IF(I4>=80,"A",IF(I4>=70,"B+",IF(I4>=60,"B",IF(I4>=50,"C+",IF (I4>=40,"C",IF(I4<=39,"NULL")))))))

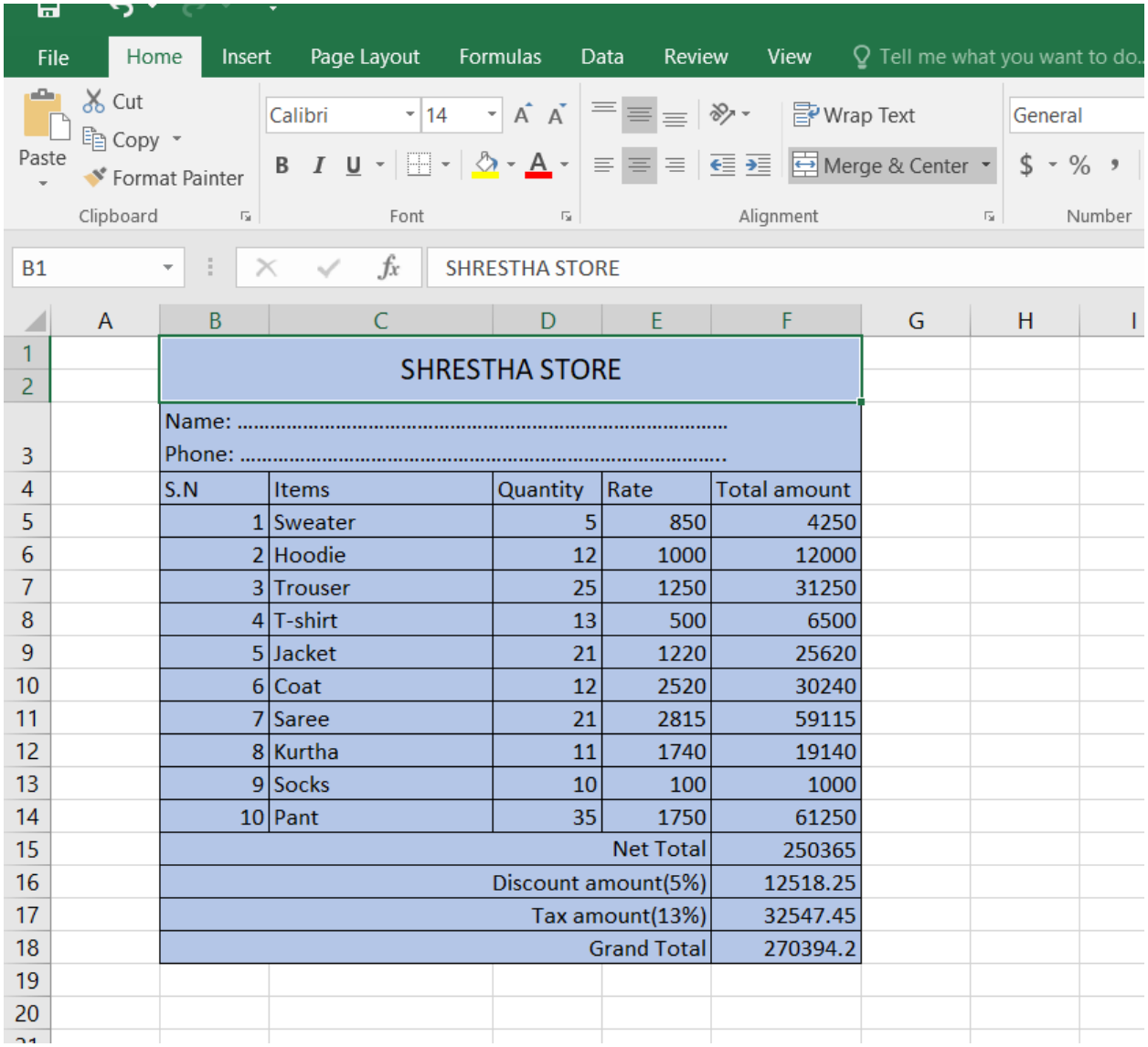
Result: =IF(MIN(C4:G4)>=24,"PASS","FAIL")

9. Prepare a bill of at least 10 items under the given conditions:

a) Enter the quantity and rate as per your choice.

b) Calculate the total amount of each.

c) Calculate Net Total, discount amount (5%), tax amount (13%) and grand total.



SHRESTHA STORE				
Name:				
Phone:				
S.N	Items	Quantity	Rate	Total amount
1	Sweater	5	850	4250
2	Hoodie	12	1000	12000
3	Trouser	25	1250	31250
4	T-shirt	13	500	6500
5	Jacket	21	1220	25620
6	Coat	12	2520	30240
7	Saree	21	2815	59115
8	Kurtha	11	1740	19140
9	Socks	10	100	1000
10	Pant	35	1750	61250
			Net Total	250365
			Discount amount(5%)	12518.25
			Tax amount(13%)	32547.45
			Grand Total	270394.2

Formulas used:

Total amount= D5*E5

Net Total= SUM (F5:F14)

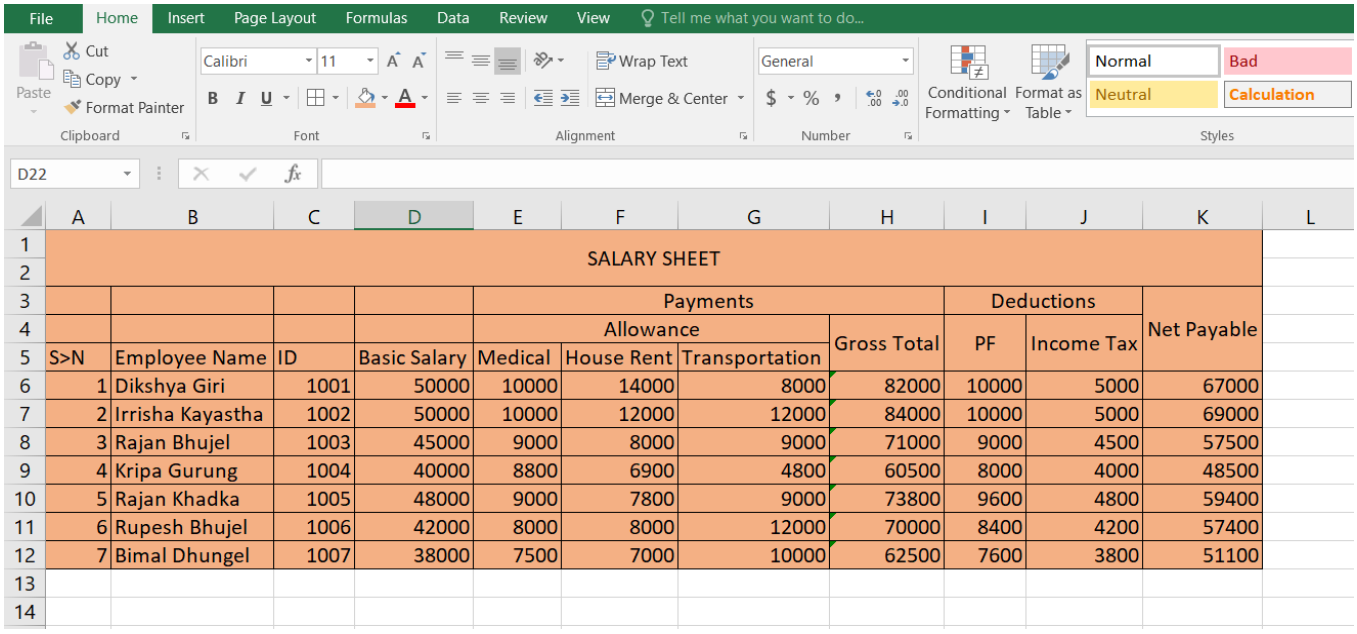
Discount amount= 5%*F15

Tax amount= 13%*F15

Grand total= F15-F16+F17

10. Create a salary sheet of at least 7 staffs in the office along with graph and chart under the following conditions:

A) Enter at least three allowances and deduct Pf @20% and Income tax @10%.



SALARY SHEET										
				Payments				Deductions		Net Payable
				Allowance			Gross Total	PF	Income Tax	
S>N	Employee Name	ID	Basic Salary	Medical	House Rent	Transportation				
1	Dikshya Giri	1001	50000	10000	14000	8000	82000	10000	5000	67000
2	Irrisha Kayastha	1002	50000	10000	12000	12000	84000	10000	5000	69000
3	Rajan Bhujel	1003	45000	9000	8000	9000	71000	9000	4500	57500
4	Kripa Gurung	1004	40000	8800	6900	4800	60500	8000	4000	48500
5	Rajan Khadka	1005	48000	9000	7800	9000	73800	9600	4800	59400
6	Rupesh Bhujel	1006	42000	8000	8000	12000	70000	8400	4200	57400
7	Bimal Dhungel	1007	38000	7500	7000	10000	62500	7600	3800	51100

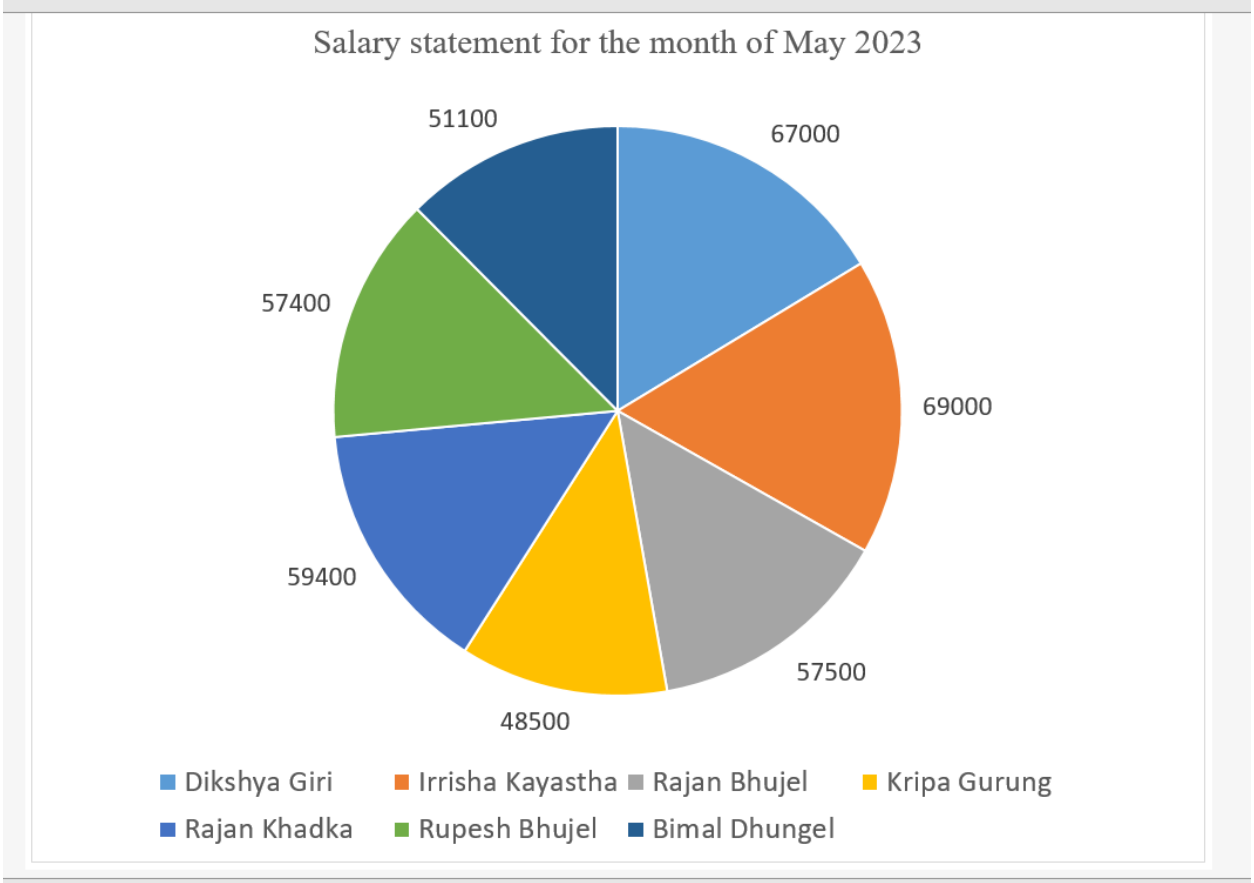
Formulas used:

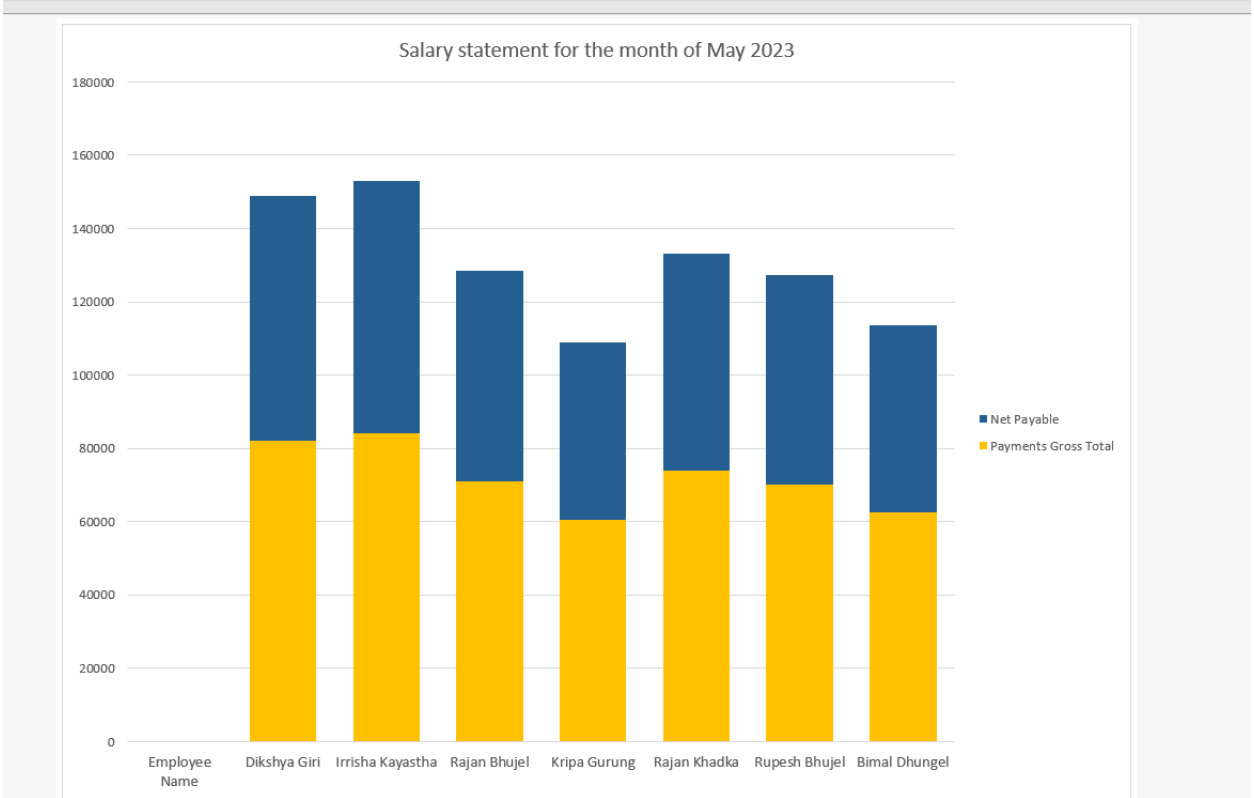
Grand Total: =SUM (D6:G6)

PF: =20%*D6

Income tax: =10%*D6

Net Payable: =H6-I6-J6


Salary statement for the month of May 2023
51100
67000
57400
69000
59400
57500
48500
Dikshya Giri
Irrisha Kayastha
Rajan Bhujel
Kripa Gurung
Rajan Khadka
Rupesh Bhujel
Bimal Dhungel


Salary statement for the month of May 2023
180000
160000
140000
120000
100000
80000
60000
40000
20000
0
Employee Name
Dikshya Giri
Irrisha Kayastha
Rajan Bhujel
Kripa Gurung
Rajan Khadka
Rupesh Bhujel
Bimal Dhungel
Net Payable
Payments Gross Total

C. Presentation software

11. Prepare a presentation including at least 10 slides of any topic of your choice related to Information Technology.

Introduction to AI

- The stimulation of human intelligence in machines that are programmed to think and act like humans,
- Used in wide range of applications like self driving cars and virtual assistants,
- Can perform tasks that require human intelligence like decision making, problem solving, learning, reasoning, etc.

Types of AI

- Narrow/ Weak AI: designed to perform specific tasks such as speech recognition, facial recognition and language recognition
- General AI: Can perform any task a human can do
- Super AI: can surpass human intelligence and can solve problems that are beyond human comprehension

"Artificial intelligence will reach human levels by around 2029. Follow that out further to, say, 2045, we will have multiplied the intelligence, the human biological machine intelligence of our civilization a billion-fold."

— Ray Kurzweil, Inventor and Futurist

Advantages of AI

- Increased Efficiency
- Cost Savings
- Improved accuracy
- Consistency
- Reduced Error

Disadvantages of AI

- Job Loss
- Potential misuse
- Over reliance on machines

THE FUTURE OF AI

- Is expected to bring greater automation in industries such as transportation, manufacturing, etc.
- Is expected to have significant impacts on healthcare, such as improving diagnostics, personalized medication and drug discovery
- Is expected to improve significantly in the upcoming years
- Advancement in self driving cars

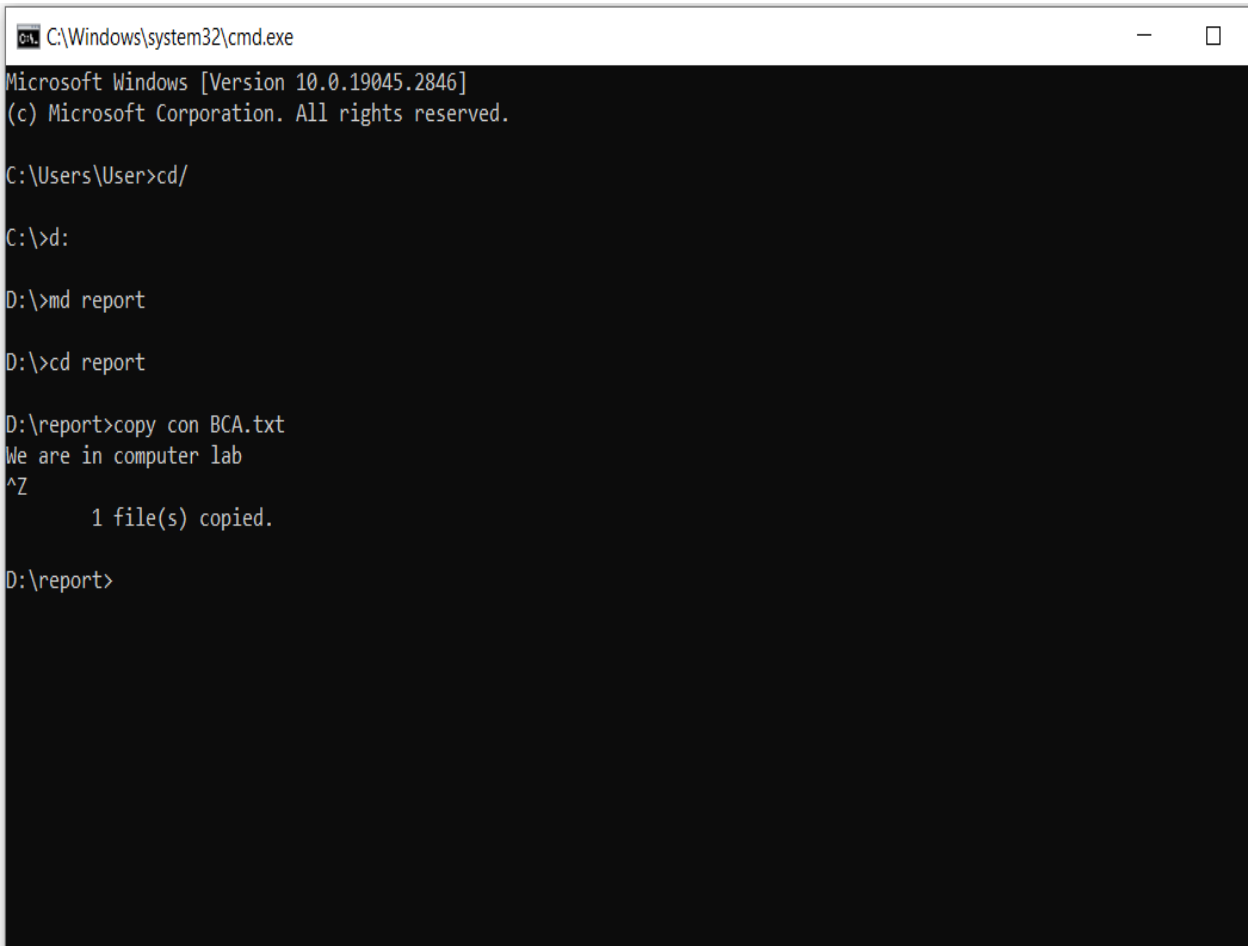
CONCLUSION

- AI is a powerful tool that can bring significant benefits for society and the economy with proper planning and consideration,
- It has the potential to drive innovation, improve productivity and enhance our quality of life.

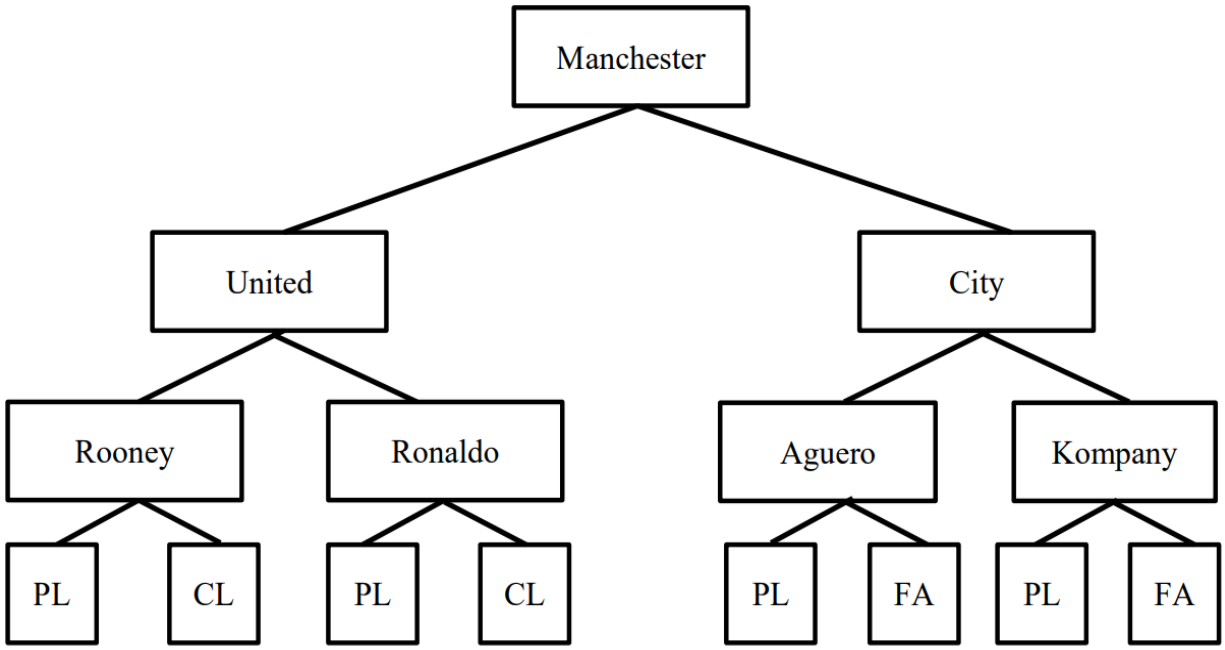
Thank you!

D. Operating system

12. Create a file named “BCA.txt” under your currently created directory. The file should contain the text “We are in computer lab”.



13. Create files and folder according to the below chart.



a. Manchester is Main Folder/Main Directory.
b. United, City, Rooney, Ronaldo, Aguero and Kompany are Sub Folder/Sub Directory.
c. PL, CL and FA are files.

```
Microsoft Windows [Version 10.0.19045.2846]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>cd/

C:\>d:

D:\>md manchester

D:\>cd manchester

D:\manchester>md united city

D:\manchester>cd united

D:\manchester\united>md Rooney Ronaldo

D:\manchester\united>cd Rooney

D:\manchester\united\Rooney>copy con PL.txt
^Z
        1 file(s) copied.

D:\manchester\united\Rooney>copy con CL.txt
^Z
        1 file(s) copied.

D:\manchester\united\Rooney>cd..

D:\manchester\united>cd Ronaldo

D:\manchester\united\Ronaldo>copy con PL.txt
^Z
        1 file(s) copied.

D:\manchester\united\Ronaldo>copy con CL.txt
^Z
        1 file(s) copied.

D:\manchester\united\Ronaldo>cd..

D:\manchester\united>cd..

D:\manchester>cd city

D:\manchester\city>md Aguero Kompany

D:\manchester\city>cd Aguero

D:\manchester\city\Aguero>copy con PL.txt
^Z
        1 file(s) copied.

D:\manchester\city\Aguero>copy con FA.txt
^Z
        1 file(s) copied.

D:\manchester\city\Aguero>cd..

D:\manchester\city>cd Kompany

D:\manchester\city\Kompany>copy con PL.txt
^Z
        1 file(s) copied.

D:\manchester\city\Kompany>copy con FA.txt
^Z
        1 file(s) copied.

D:\manchester\city\Kompany>cd/

D:\>
```